RESEARCH METHODS / PRACTICAL FIELD GUIDE

Survey-Based Quantitative Research

A step-by-step guide to questionnaire validation, data quality, hypothesis testing, and reporting

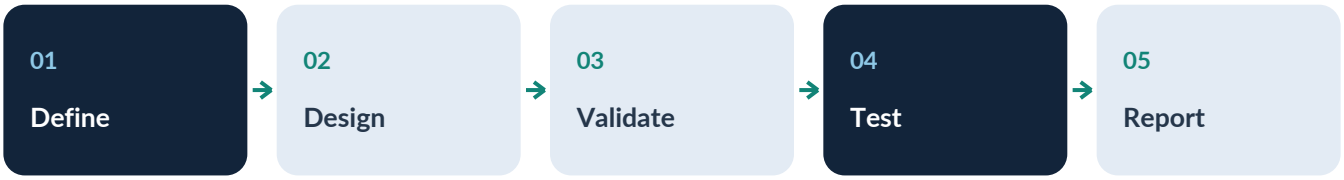
A reusable framework for studies in business, finance, technology, education, and other social sciences. The AI-agent loan appraisal example is illustrative only.

WHAT THIS GUIDE COVERS

Research model | Survey design | Reliability | EFA / CFA | Regression / SEM | Reporting

01 Research roadmap

This guide distinguishes two different questions: **Is the survey instrument credible?** and **Do the data support the research hypotheses?** Both must be addressed before drawing conclusions.



1.1 Suggested paper structure

Section	Primary output
1. Introduction	Problem, research gap, objectives, research questions, contribution.
2. Literature review	Theories, construct definitions, conceptual model, hypotheses.
3. Methodology	Population, sampling, instrument development, ethics, analysis plan.
4. Data and measurement validation	Screening, respondent profile, reliability, EFA / CFA, validity.
5. Hypothesis testing	Assumption checks, models, effect sizes, confidence intervals.
6. Discussion and conclusion	Interpretation, implications, limitations, future research.

Two kinds of evidence

Measurement tests examine whether questionnaire items represent the intended constructs. Hypothesis tests examine associations or group differences involving those constructs. A statistically significant regression cannot rescue an invalid measure.

1.2 Study design decision

- **Cross-sectional:** survey participants once; appropriate for perceptions and associations, not strong causal claims.
- **Longitudinal:** survey repeatedly to observe changes over time; still requires careful causal identification.
- **Intervention or quasi-experiment:** use baseline and follow-up measurements, ideally with a comparison group, for stronger evidence about actual effects.

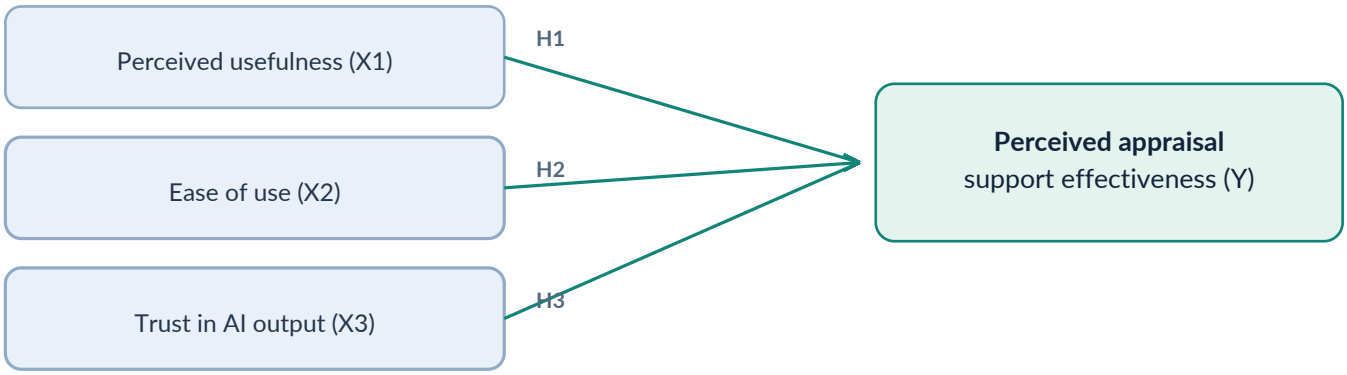
02 Theory, variables, and hypotheses

Begin with a precise research question. Define each concept in measurable terms before drafting individual questions.

2.1 Specify constructs

Variable role	Purpose / example
Independent (X)	Predictor, such as perceived usefulness of AI agents.
Dependent (Y)	Outcome, such as self-reported effectiveness of appraisal support.
Mediator (M)	Potential mechanism, such as reduced cognitive workload.
Moderator (W)	Condition changing an effect, such as AI experience.
Control (C)	Relevant characteristics, such as role or tenure, selected theoretically.

2.2 Illustrative model - not a validated banking instrument



Example hypotheses: **H1**: Perceived usefulness is positively associated with perceived effectiveness. **H2**: Ease of use is positively associated with perceived effectiveness. **H3**: Trust in AI output is positively associated with perceived effectiveness.

Specify the outcome carefully

A survey of perceptions can test **perceived effectiveness** or **intention to use**. It cannot, by itself, establish improvement in objective loan quality, default risk, or processing time. Those outcomes require operational or experimental data.

03 Questionnaire and sampling design

3.1 Build a measurement blueprint

For each construct, document its definition, source scale, proposed items, response anchors, and reverse-coded items. Adapt established instruments with attribution rather than inventing every question.

Construct	Illustrative item	Scale
Perceived usefulness	"Using an AI agent would help me complete relevant appraisal tasks more efficiently."	1-5 agreement
Trust in AI output	"I would verify AI-generated recommendations before using them in an appraisal decision."	1-5 agreement
Perceived effectiveness	"AI-agent assistance could improve the consistency of my appraisal workflow."	1-5 agreement

Item-writing rules: use one idea per item; avoid leading, vague, absolute, and double-barreled statements; make response anchors explicit; keep timeframes consistent; separate demographic questions from psychological constructs.

3.2 Sampling plan

- Define the target population, inclusion/exclusion criteria, unit of analysis, and recruitment channel.
- Select probability sampling where feasible; if using convenience/purposive sampling, explain generalizability limits.
- Determine sample size **before** data collection using power analysis and expected effect size; also consider the number of factors/items and model complexity.
- Document approached, eligible, consenting, completed, and usable respondents. Report response rate only when the denominator is credible.

3.3 Pilot test and research ethics

- Seek expert review for relevance, clarity, completeness, and translation equivalence when applicable.
- Pilot with representatives of the target population; a pilot of roughly 20-30 can identify wording issues but is not a substitute for full psychometric validation.
- Use a clear consent statement: voluntary participation, confidentiality, data retention, and who will access the data.

Good practice

Predefine your constructs, hypotheses, primary outcomes, exclusion rules, and main statistical method before analyzing the main survey. A short analysis plan reduces post hoc decision-making.

04 Data quality and screening

Establish an audit trail: preserve the raw dataset, create a cleaned analytical dataset, and document every recoding or exclusion.

4.1 Screening checklist

Check	Practical action
Eligibility and duplicates	Remove ineligible cases; identify duplicate responses using defensible criteria.
Missing values	Report missingness per item/person; assess patterns and justify complete-case analysis or imputation.
Speeding / straight-lining	Flag implausibly fast completions, repetitive response patterns, and failed attention checks; avoid automatic deletions without context.
Reverse coding	Reverse designated items before constructing scale scores; document the scoring rule.
Outliers and impossible values	Check ranges, influential cases, and data entry problems; distinguish errors from true extremes.
Representativeness	Compare available respondent characteristics with population or recruitment records; examine possible nonresponse bias.

4.2 Descriptive overview

- Report n, demographic frequencies and percentages; for constructs, report mean, standard deviation, minimum, maximum, and response distribution.
- Inspect floor and ceiling effects and item distributions. For ordinal Likert items, consider ordinal-appropriate estimation where necessary.
- Describe missing-data handling explicitly; do not silently drop all incomplete cases without checking whether this could bias results.

4.3 Recommended dataset fields

Keep an anonymous respondent ID, survey wave/date (if applicable), each item as a separate column, reverse-scored variables, construct scores, group variables, and an exclusion-log flag. Never publish identifiable personnel or customer information.

Deliverable

A **data screening table** with initial n, exclusions by reason, final analytical n, missingness summary, and descriptive statistics.

05 Reliability and exploratory factors

5.1 Internal consistency

Reliability addresses whether items within a proposed reflective construct behave consistently. It does not prove the construct is valid. Evaluate results alongside item wording and content coverage.

Statistic	Common guideline	Interpretation
Cronbach's alpha	About 0.70+	A conventional benchmark; affected by item count and assumptions.
McDonald's omega	About 0.70+	Often preferable when item loadings differ.
Corrected item-total correlation	About 0.30+	Low values signal potentially weak or miscoded items.
Composite reliability	About 0.70+	Used in latent-variable measurement models.

Avoid deleting items solely to increase alpha. Check conceptual coverage, negative wording, redundancy, dimensionality, and any reverse-coding errors first.

5.2 Exploratory Factor Analysis (EFA)

- **Use EFA when:** the dimensional structure is unknown, substantially adapted, or needs exploration.
- **Check suitability:** inspect the correlation matrix, sample adequacy (KMO commonly at least 0.60), and Bartlett's sphericity test (often $p < 0.05$).
- **Extract and rotate:** use common-factor methods and oblique rotation when factors may correlate. Consider parallel analysis and interpretability when deciding how many factors to keep.
- **Review items:** assess factor loadings (often around 0.40-0.50+), cross-loadings, and the conceptual meaning of each factor.
- **Reconsider the model:** if factor structure differs substantially, revise it transparently; revalidate on an independent or held-out sample when possible.

EFA is not always mandatory

A well-established scale with strong theoretical justification can proceed directly to CFA. If both EFA and CFA are performed, avoid treating confirmation on the same exploratory sample as independent validation.

06 Confirmatory measurement validation

6.1 Confirmatory Factor Analysis (CFA)

CFA tests whether observed indicators fit a prespecified latent-variable model. It should be evaluated using theory, standardized loadings, residuals, fit indices, and identification checks, rather than a single threshold.

Evidence	Indicative convention	Notes
Standardized loading	Preferably 0.50+; often 0.70+	Review weak items in context rather than blindly deleting them.
Composite reliability (CR)	0.70+	Internal consistency within the latent construct.
Average variance extracted (AVE)	0.50+	Convergent validity, jointly interpreted with loadings.
HTMT ratio	Below 0.85 or 0.90	Discriminant validity; report method/criterion used.
CFI / TLI	Around 0.90+	Global model fit; interpretation depends on the model.
RMSEA / SRMR	Often below 0.08	Approximate fit; include intervals where relevant.

6.2 Estimation and validity decisions

- Use an estimator appropriate for the data; ordinal CFA with polychoric correlations and robust weighted least squares may suit Likert items with few response categories.
- Do not repeatedly modify a model based only on modification indices. Any correlated errors or item removal must be theoretically justified and disclosed.
- If comparing demographic groups, evaluate measurement invariance before interpreting mean differences across latent constructs.

6.3 Common method bias (CMB)

When predictors and outcomes are collected from the same respondent using one survey, consider procedural controls: separate question blocks, neutral wording, anonymity, and different data sources or waves if feasible. Post hoc diagnostics can support assessment but do not establish the absence of CMB. Harman's single-factor test alone is insufficient.

Reflective vs. formative

The alpha / omega / AVE / CFA logic assumes reflective measurement. Formative composites require a different validity strategy. Specify construct type before applying a checklist.

07 Select the statistical test

The right statistical method follows the research question, outcome type, study design, and measurement level - not a preference for one software package.

Research question	Typical analysis	Key requirement
Is the average score different from a reference?	One-sample t-test	Meaningful comparison value; assumptions checked.
Do two independent groups differ?	t-test; Mann-Whitney U	Group independence; consider Welch's t-test.
Do three or more groups differ?	ANOVA / Welch ANOVA; Kruskal-Wallis	Post hoc contrasts and multiple-testing control.
Are two variables associated?	Pearson or Spearman correlation	Choose based on construct/scale and distribution.
Do multiple X variables predict a numeric Y?	Multiple regression	Diagnostics for functional form and errors.
Is the dependent variable binary?	Logistic regression	Sufficient events and appropriate specification.
Are relationships among latent constructs tested?	CB-SEM / PLS-SEM	Choose based on inferential goal and model type.
Does an association pass through M or change with W?	Mediation / moderation analysis	Theory, temporal order and uncertainty estimates.

7.1 Assumption testing for regression

- Check linearity, independence or clustering of observations, influential cases, and possible omitted-variable concerns.
- Examine residual patterns for heteroscedasticity and non-normality when relevant; use robust standard errors where justified.
- Inspect predictor collinearity (for example, VIF); do not use a single cutoff as a substitute for interpretation.

7.2 Choose one primary analysis

Identify the main estimand and a prespecified primary model. Treat subgroup and robustness analyses as secondary, and account for multiplicity where many hypotheses are tested.

08 Hypothesis testing and interpretation

8.1 Recommended reporting elements

- Report the estimated coefficient or group difference, direction, standard error (if applicable), **95% confidence interval**, exact p-value, and an effect-size measure.
- For regression, report n, coding, standardized/unstandardized coefficients as appropriate, R-squared / adjusted R-squared, and diagnostics.
- For SEM, report measurement model results, structural paths, interval estimates, model fit or predictive metrics, and estimator / bootstrap settings.
- State how missing values, influential observations, alternative specifications, and multiple comparisons were handled.

8.2 Hypothesis summary template

Hypothesis	Effect / path	Estimate	95% CI	p-value	Decision
H1	X1 to Y	[...]	[...]	[...]	[supported?]
H2	X2 to Y	[...]	[...]	[...]	[supported?]
H3	X3 to Y	[...]	[...]	[...]	[supported?]

8.3 A model reporting paragraph

"In the adjusted model, perceived usefulness was positively associated with perceived effectiveness (beta = [estimate], 95% CI [lower, upper], p = [value]). The model explained [R-squared] of the observed outcome variance. The association remained [similar/different] in prespecified sensitivity analyses."

8.4 Statistical interpretation pitfalls

- **p < 0.05 is not the whole result:** distinguish statistical evidence from practical importance and uncertainty.
- **Non-significant is not zero:** evaluate precision and plausible effect sizes, not just a binary label.
- **Correlation is not causation:** avoid "AI agents improve loan appraisal" when measuring self-reported perceptions at one time point.
- **Model selection bias:** disclose exploratory models and post hoc changes instead of presenting them as prespecified.

Illustrative wording

Prefer "greater trust was associated with higher perceived effectiveness" over "trust caused appraisal performance to improve" in a cross-sectional survey.

09 How to write the paper

Section 1 - Introduction

Explain the operational problem, significance, empirical gap, research objectives and questions, scope, and expected contributions.

Section 2 - Literature review and hypotheses

Define all constructs; summarize key theories and empirical findings; derive a conceptual framework and directional hypotheses.

Section 3 - Methodology

Describe study design, population, sampling strategy, sample-size rationale, survey development, sources of measurement items, pilot findings, ethical safeguards, data collection, and analysis plan.

Section 4 - Data quality and measurement results

Provide a respondent flow, demographics, missing-data decisions, descriptive statistics, reliability, dimensionality, measurement model fit, convergent/discriminant validity, and common-method-bias safeguards.

Section 5 - Hypothesis testing results

Present bivariate results only as context; focus on the primary model, relevant diagnostics, estimates, confidence intervals, effect sizes, and robustness checks. Summarize each hypothesis without overstating evidence.

Section 6 - Discussion and conclusion

Interpret substantive findings against prior work. Separate theoretical implications from management implications. Explain design limitations, generalizability concerns, future research, and the key conclusion.

Suggested tables and figures

Figure 1: conceptual research model. Table 1: construct definitions and measurement items. Table 2: sample characteristics. Table 3: reliability and validity. Table 4: correlation matrix. Table 5: primary regression or SEM results. Table 6: hypothesis summary.

What belongs in the appendix?

Full questionnaire and response anchors; exact item sources and translation notes; consent text; analysis protocol; diagnostic plots; detailed EFA/CFA loadings; codebook; and supplementary analyses.

10 End-to-end quality checklist

Use this checklist before submitting a report, thesis chapter, or journal manuscript.

Phase	Minimum deliverable / check
Planning	Clearly framed research question; theory-backed constructs and hypotheses; ethical approval/consent where needed.
Instrument	Item-source table, construct definitions, response anchors, expert review, pilot feedback, revision history.
Sampling	Defined population, recruitment procedure, sample-size rationale, eligibility and usable-response counts.
Cleaning	Raw-to-final response flow, missingness handling, exclusions and reverse-scoring documented.
Measurement	Reliability evidence; EFA if warranted; CFA / appropriate construct-validity checks if warranted.
Bias	Assessed sampling, nonresponse, common-method and same-source concerns; limitations stated.
Inference	Suitable test selected; assumptions examined; estimates, confidence intervals, p-values, effect sizes reported.
Discussion	Results compared with literature; practical meaning separated from statistical significance.
Transparency	Questionnaire, item codebook, software/version, model specifications, reproducible analysis where possible.

A compact analysis sequence

1. Design and pilot the survey → **2.** Collect and screen responses → **3.** Establish measurement quality → **4.** Run prespecified statistical tests → **5.** Interpret limitations and report transparently.

Minimum standard for defensible conclusions

The survey must measure the intended concepts; the sample must be described honestly; the statistical model must match the data; and claims must not exceed what the research design can establish.

11 Selected methodology references

These sources provide further guidance. Numeric cutoffs in the preceding pages are heuristics, not universal pass/fail rules.

Measurement and reliability

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, *16*, 297-334.

McDonald, R. P. (1999). *Test Theory: A Unified Treatment*. Lawrence Erlbaum Associates.

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*, 115-135.

Model fit, bias, and reporting

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, *6*(1), 1-55.

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, *88*(5), 879-903.

American Psychological Association. (2020). *Publication Manual of the American Psychological Association* (7th ed.).

Scope of this guide

This is a general workflow for survey-based quantitative research, not a claim that every study must conduct every test. Choose EFA, CFA, regression, SEM and sensitivity analyses according to the construct model, available data, design, and research objective.

END OF GUIDE